

Forder — Book IV

Proposition 15

Formal Reconstruction — Technical Documentation

Jurek Rózański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJTEAMLAB

If $ABCD$ is a cyclic quadrilateral, then the opposite angles BAD and BCD are supplementary.

1 Introduction

Forder IV.15 is the general cyclic-quadrilateral supplement theorem: if A, B, C, D form a cyclic quadrilateral, then the opposite angles

$$\angle BAD \quad \text{and} \quad \angle BCD$$

are supplementary [1].

In the source this proposition is proved from IV.12.1, IV.13, and IV.14 by introducing the centre of the circumcircle and, in the generic case, an antipodal point. The formal reconstruction follows exactly that geometric strategy, but it makes explicit several order facts that Forder leaves in the diagram.

The production file is

```
CGJteamLab/Proposition4_15.lean
```

and the public theorem is

```
hilbert_IV15_cyclic_supplement
```

IV.15 is also the first proposition in this Book IV block whose formal proof uses all three earlier numbered results:

$$\text{IV.12.1,} \quad \text{IV.13,} \quad \text{IV.14.}$$

2 Classical Configuration

Let A, B, C, D be four points on one circle. We use the diagonal BD as the reference carrier. In the convex cyclic configuration the other two vertices A, C lie on opposite sides of BD .

A, C opposite sides of BD

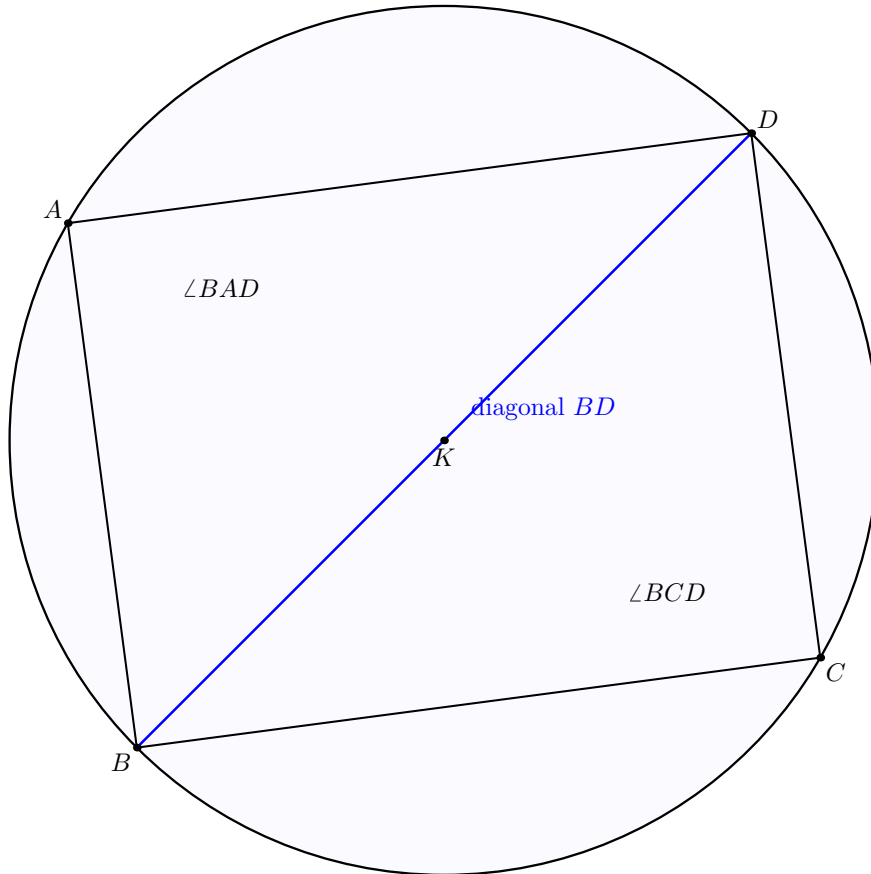


Figure 1: Forder IV.15. The cyclic quadrilateral is organized around the diagonal BD . The vertices A, C lie on opposite sides of that diagonal, and the target is that $\angle BAD$ and $\angle BCD$ are supplementary.

The production interface does not leave this convexity information implicit. It asks for an explicit line `diagonal` through B, D and the hypothesis

```
HilbertOppositeSide Geo A C diagonal
```

This is the exact order information needed by the proof.

3 Forder's Statement and Proof

Forder states:

If $ABCD$ is a cyclic quadrilateral, then $\angle BAD + \angle BCD = \pi$.

He first notes that A and C lie on opposite sides of the diagonal BD . Let O be the centre of the circle.

If O lies on BD , then BD is a diameter. Both $\angle BAD$ and $\angle BCD$ are right by IV.13, so they are supplementary.

Suppose now that O is off BD . Since A, C lie on opposite sides, the centre lies on the same side of BD as exactly one of them. Forder chooses the case in which O is on the same side as C . Let E be the point of the circle antipodal to A , so that

$$A - O - E.$$

The diameter AE crosses the diagonal BD , and the relevant order relations imply that B, D lie on opposite sides of AE . IV.14 applied to the cyclic quadrilateral A, B, E, D gives

$$\angle BAD + \angle BED = \pi.$$

On the other hand C, O, E lie on the same side of BD , so IV.12.1 gives

$$\angle BED \cong \angle BCD.$$

Substitution yields

$$\angle BAD + \angle BCD = \pi.$$

The printed proof handles the alternative placement of the centre by symmetry. The formal proof makes that symmetry explicit by relabelling the cyclic quadrilateral.

4 Formal Statement

The public proposition contract is

```
def HilbertForderIV15CyclicSupplement
  [H : HilbertIncidence Geo]
  [_H0 : @HilbertOrder Geo H] : Prop :=
  forall A B C D : Geo.Point,
    forall diagonal : Geo.Line,
      Ne B D ->
        H.OnLine B diagonal ->
          H.OnLine D diagonal ->
            HilbertConcyclic4 Geo A B C D ->
              HilbertOppositeSide Geo A C diagonal ->
                Not (PrimCollinear Geo B A D) ->
                  Not (PrimCollinear Geo B C D) ->
                    exists X : Geo.Point,
                      BookZeroSupplement Geo
                        B C D
                        D X /\
                          Geo.AngleCongruent
                            B A D
                            X C D
```

The theorem therefore states the supplement synthetically. The witness X satisfies the straight-line extension condition associated with $\angle BCD$, and the angle

$$\angle XCD$$

is congruent to the opposite cyclic angle

$$\angle BAD.$$

The closed production theorem is

```
theorem hilbert_IV15_cyclic_supplement
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV15CyclicSupplement (Geo := Geo)
```

5 Recovering an Explicit Circumcircle

The hypothesis

```
HilbertConcyclic4 Geo A B C D
```

does not expose the centre directly. The first step of the production proof is therefore

```
rcases
  hilbert_concyclic4_circle
    Geo A B C D hCyclic
  with
    <K, hAcircle, hBcircle, hCcircle, hDcircle>
```

This produces an explicit centre K and circle membership witnesses for all four vertices.

The rest of the proof is a classification of the position of K relative to the diagonal BD .

6 Case I: the Centre Lies on the Diagonal

Assume

$$K \in BD.$$

Then BD is a diameter of the circumcircle.

IV.13 gives

$$\angle BAD \text{ right, } \angle BCD \text{ right.}$$

The formal proof extends BC through C to X :

$$B - C - X.$$

The right angle at C is transported across the straight carrier so that

$$\angle XCD$$

is also right. Since all right angles are congruent,

$$\angle BAD \cong \angle XCD.$$

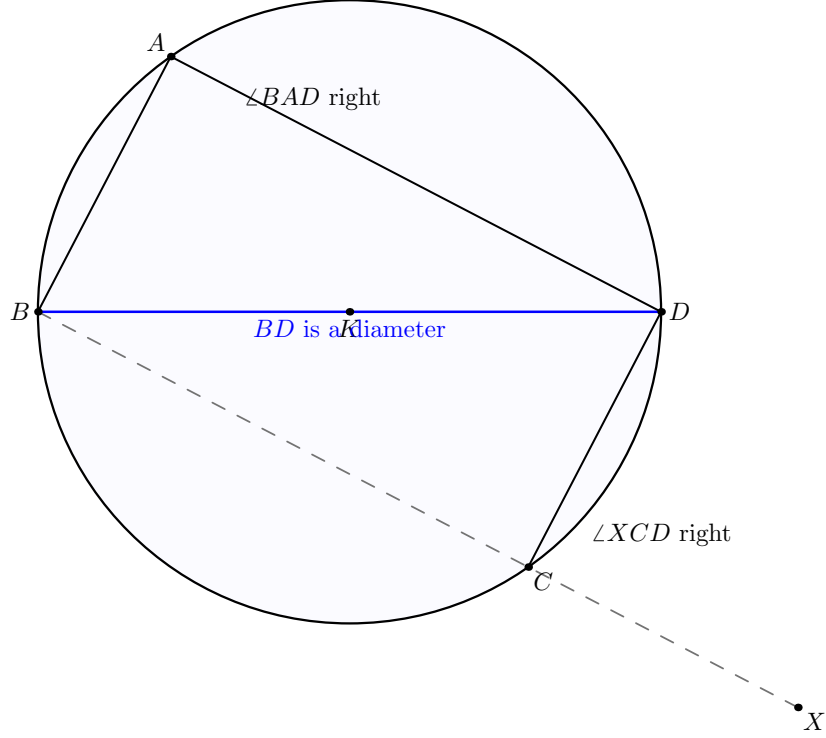


Figure 2: The exceptional branch $K \in BD$. The diagonal BD is a diameter, so both opposite angles BAD and BCD are right by IV.13.

The point X therefore witnesses

$$\text{Supplement}(BCD, DCX),$$

and the IV.15 conclusion follows immediately.

This branch is implemented by

```
hilbert_forder_IV15_center_on_diagonal_stage44
```

and depends essentially only on IV.13 and the Book Zero notion of a supplement.

7 The Generic Case: the Centre Is Off the Diagonal

Assume now

$$K \notin BD.$$

Since A and C lie on opposite sides of BD , exactly one of the two vertices is on the same side of the diagonal as K .

The proof first treats the branch

$$C \sim K \text{ relative to } BD.$$

This is the branch that corresponds directly to Forder's printed diagram.

8 Constructing the Antipode of A

Let E be the antipode of A :

$$A - K - E,$$

with E on the same circle.

The reusable theorem is

`hilbert_circle_antipode`

The line AE is therefore a diameter.

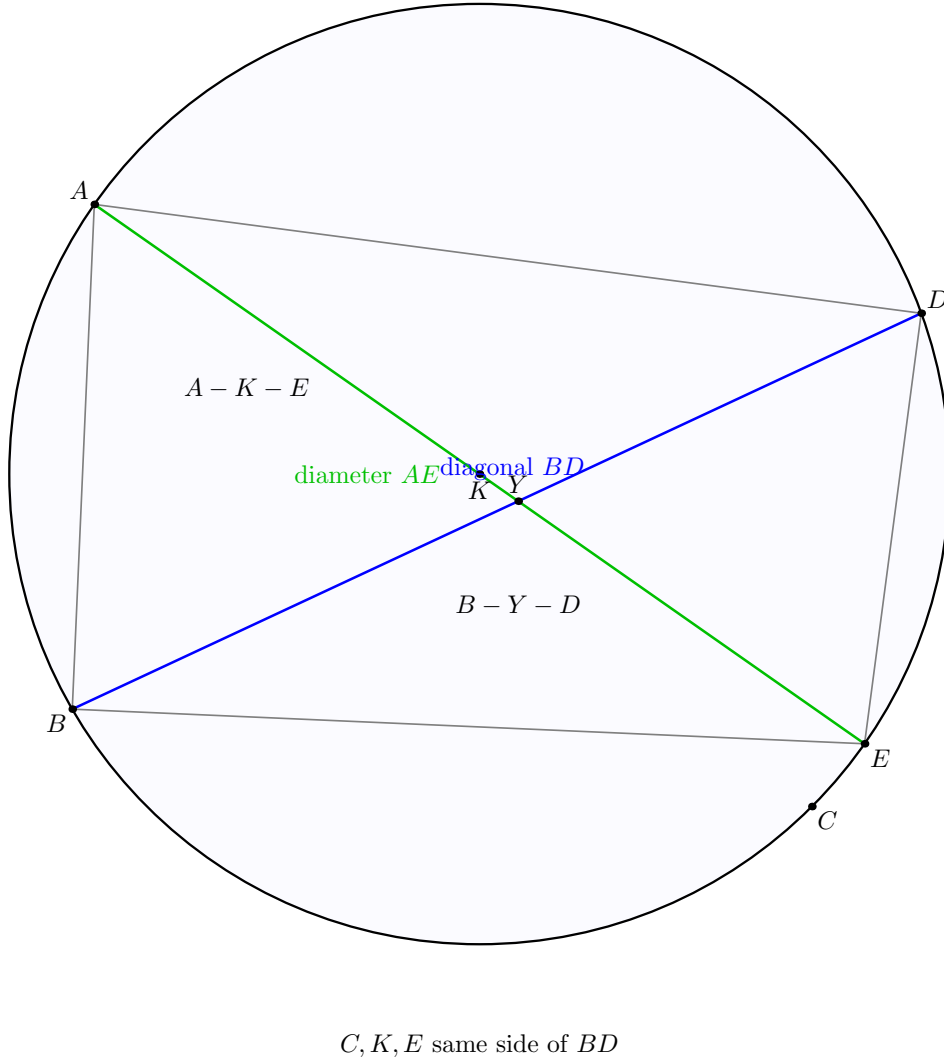


Figure 3: Generic IV.15 branch. The centre K lies on the same side of the diagonal BD as C . The point E is the antipode of A , so $A - K - E$. The diameter AE crosses BD at Y .

The formal proof next establishes that A and E lie on opposite sides of the diagonal BD . The argument is pure order transport.

Because A, C are opposite sides and C, K are same side, A, K are opposite sides. Let Y be their crossing point on BD :

$$A - Y - K.$$

Since also

$$A - K - E,$$

betweenness transitivity gives

$$A - Y - E.$$

Thus the same point Y witnesses that A, E lie on opposite sides of BD .

The proposition-specific helper is

```
hilbert_IV15_antipode_opposite_diagonal_stage45
```

9 Where Is the Intersection Point on the Diameter?

The crossing point Y lies on both the diagonal BD and the diameter AE . To control the position of B, D relative to AE , the proof first needs to know that Y lies inside the circle unless it is the centre.

From

$$A - K - E, \quad A - Y - E, \quad KA \cong KE,$$

the local lemma

```
hilbert_IV15_diameter_inner_point_stage46
```

proves

$$Y = K \quad \text{or} \quad KY < KA.$$

The proof is a trichotomy on the collinear points A, K, Y .

If $A - K - Y$, then $K - Y - E$, so

$$KY < KE \cong KA.$$

The order $K - A - Y$ is incompatible with $A - K - E$ and $A - Y - E$.

If $A - Y - K$, then $K - Y - A$ directly gives

$$KY < KA.$$

Thus the intersection point is either the centre or strictly inside the circle.

10 Why B, D Must Be on Opposite Sides of the New Diameter

The key order theorem of the generic branch is that the chord endpoints B, D lie on opposite sides of the diameter AE .

The difficulty is subtle. We know only that Y lies on the chord carrier BD and on the diameter AE . To prove opposite-side status we must show

$$B - Y - D.$$

The points B, Y, D are collinear, so order trichotomy gives three possibilities:

$$B - Y - D, \quad Y - B - D, \quad B - D - Y.$$

The first is the desired order.

The other two would put Y outside the chord. Since

$$KB \cong KD,$$

the two exterior-order lemmas

```
hilbert_IV15_chord_exterior_farther_left_stage47
hilbert_IV15_chord_exterior_farther_right_stage47
```

show respectively that an exterior Y would satisfy

$$KD < KY \quad \text{or} \quad KB < KY.$$

But the previous diameter lemma gives

$$Y = K \quad \text{or} \quad KY < KA,$$

while all circle radii satisfy

$$KA \cong KB \cong KD.$$

Both exterior orders therefore contradict the radius comparison.

Hence only

$$B - Y - D$$

is possible, and the diameter AE genuinely separates B from D .

The proposition-specific theorem is

```
hilbert_IV15_chord_opposite_diameter_stage48
```

This is one of the places where the formal proof contains much more order geometry than the source text: Forder's figure treats this separation as visually evident.

11 Applying IV.14 to the Antipodal Quadrilateral

Now A, E are endpoints of a diameter, while B, D lie on opposite sides of that diameter. IV.14 applies to the four circle points

$$A, B, E, D.$$

It constructs a point X such that

$$\angle BAD \cong \angle XED$$

and $\angle XED$ is a supplement of

$$\angle BED.$$

In source notation this is exactly

$$\angle BAD + \angle BED = \pi.$$

This is the first half of Forder's generic argument.

12 Transporting the Same-Segment Angle

We next compare $\angle BED$ with the original angle $\angle BCD$.

From A, K being opposite sides of BD and the order

$$A - K - E,$$

the point E lies on the same side of BD as K .

Since the current branch assumes that C, K are also on the same side of BD , both C and E are in the same segment determined by chord BD .

IV.12.1 therefore gives

$$\angle BED \cong \angle BCD.$$

The production proof uses

```
hilbert_IV12_same_segment
```

directly.

The same-side transport from $A - K - E$ is now supplied by the reusable interface theorem

```
hilbert_oppositeSide_tail_sameSide
```

rather than by a proposition-specific stage helper.

13 Transporting Supplements

The proof now has:

$$\angle BAD \cong \text{a supplement of } \angle BED,$$

and

$$\angle BED \cong \angle BCD.$$

The target, however, is not merely to say that congruent angles have equal numerical supplements. We must construct the actual supplement at C .

Extend BC through C to X_c :

$$B - C - X_c.$$

This supplies

```
BookZeroSupplement Geo B C D D Xc
```

Book Zero theorem 43 on supplements then transports the congruence

$$\angle BED \cong \angle BCD$$

to a congruence of their supplements. After reorienting the angle arms, the formal chain is

$$\angle BAD \cong \angle XED \cong \angle X_c CD.$$

Therefore

$$\boxed{\angle BAD \cong \angle X_c CD}.$$

This completes the branch $C \sim K$.

The proposition-specific theorem is

```
hilbert_forder_IV15_sameSide_CK_stage50
```

14 The Other Side of the Diagonal: Explicit Symmetry

The source proof simply chooses the case in which the centre is on the same side of BD as C . Formally, when this is false we must close the alternative branch.

Assume

$$C \not\sim K.$$

Because both are off the diagonal, they are opposite sides:

$$C \mid K.$$

Since $A \mid C$, the two points A, K are on the same side of BD .

The reusable theorem

```
hilbert_two_opposites_sameSide_with_carrier
```

performs this plane-separation inference.

Now apply the already proved same-side- K branch after the cyclic relabelling

$$(A, B, C, D) \mapsto (C, D, A, B).$$

The relabelled theorem returns a supplement relation based at A and an angle congruence

$$\angle DCB \cong \angle XAB.$$

After reversing the first angle this is

$$\angle BCD \cong \angle XAB.$$

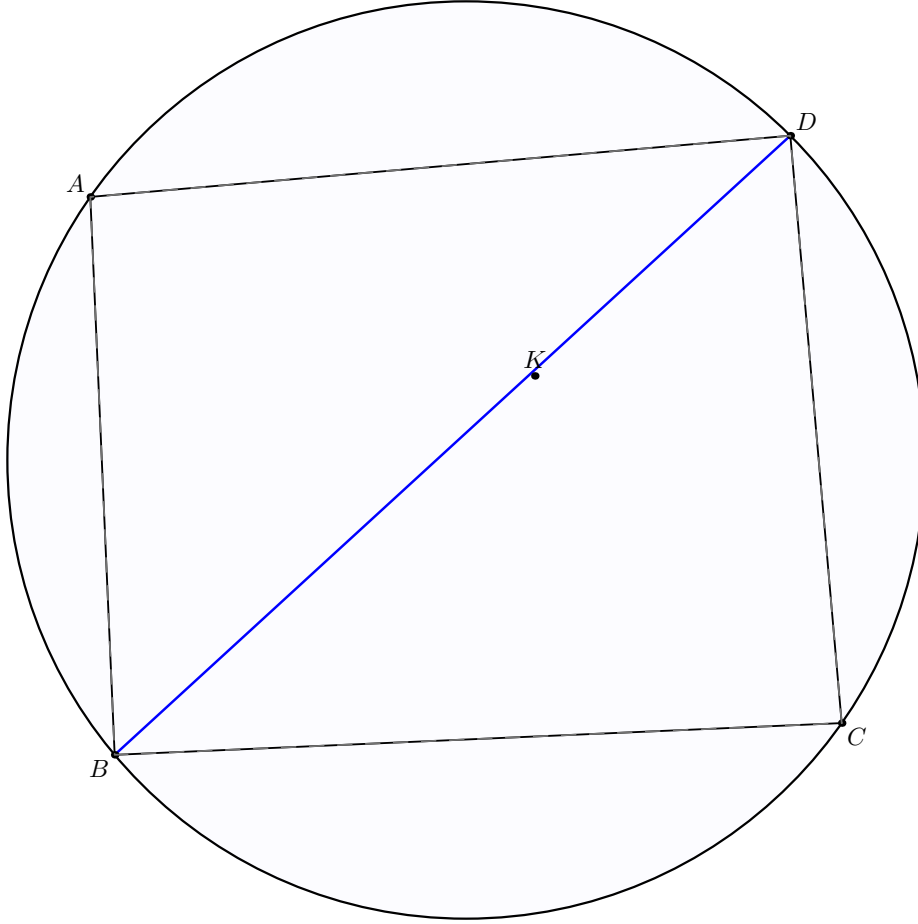
The same witness also shows that $\angle BAD$ is the supplement of $\angle XAB$. The proof then constructs the target supplement of $\angle BCD$ at C and invokes Book Zero supplement transport once more.

The result is

$$\angle BAD \cong \angle X_c CD.$$

This explicit relabelling is the formal replacement for the phrase “the other case is symmetric”.

K on the A -side of BD



apply completed branch to

$$(A, B, C, D) \mapsto (C, D, A, B)$$

Figure 4: The branch not drawn by Forder. If K lies on the same side of BD as A rather than C , the proof applies the completed generic branch to the cyclic relabelling $(A, B, C, D) \mapsto (C, D, A, B)$, then transports the resulting supplement back to the original orientation.

15 Completion of IV.15

The public proof has three branches.

1. $K \in BD$: the diagonal is a diameter, so both target angles are right by IV.13.
2. $K \notin BD$ and C, K are on the same side: construct the antipode of A , apply IV.14, then IV.12.1, and transport supplements.
3. $K \notin BD$ and C, K are not on the same side: prove A, K same side, cyclically relabel the quadrilateral, apply the completed generic branch, and transport the result back.

This case split is implemented by

```
hilbert_forder_IV15_complete_stage52
```

and the public theorem is the thin production wrapper

```
theorem hilbert_IV15_cyclic_supplement
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV15CyclicSupplement (Geo := Geo) :=
  hilbert_forder_IV15_complete_stage52 Geo
```

16 Lean Formalization

The proposition-specific implementation layer consists of:

```
hilbert_forder_IV15_center_on_diagonal_stage44
hilbert_IV15_antipode_opposite_diagonal_stage45
hilbert_IV15_diameter_inner_point_stage46
hilbert_IV15_chord_exterior_farther_left_stage47
hilbert_IV15_chord_exterior_farther_right_stage47
hilbert_IV15_chord_opposite_diameter_stage48
hilbert_forder_IV15_sameSide_CK_stage50
hilbert_forder_IV15_complete_stage52
```

These names remain private implementation details of `Proposition4_15.lean`.

The generic infrastructure used by those helpers includes:

```
hilbert_concyclic4_circle
hilbert_circle_antipode
hilbert_circle_center_midpoint_of_chord
hilbert_sameRay_beyond_common_middle
hilbert_oppositeSide_tail_sameSide
hilbert_two_opposites_sameSide_with_carrier
bookZero_43_supplements
```

The numbered dependencies are explicit:

```
hilbert_IV12_same_segment
hilbert_IV13_diameter_right_angle
hilbert_IV14_diameter_supplement
```

This is the intended production architecture: proposition-specific order lemmas remain local, while general side-of-line and circle facts are owned by the interface.

17 Relation to IV.12.1, IV.13, and IV.14

IV.15 is where the preceding circle results first assemble into a single theorem.

17.1 Use of IV.13

If the centre lies on diagonal BD , IV.13 closes the theorem immediately: both opposite angles stand on the diameter BD .

17.2 Use of IV.14

In the generic branch the antipodal diameter AE is introduced precisely so that IV.14 can be applied to A, B, E, D .

17.3 Use of IV.12.1

Once E is shown to lie in the same segment of chord BD as C , IV.12.1 replaces the auxiliary angle

$$\angle BED$$

by the original target angle

$$\angle BCD.$$

Thus the classical proof architecture is exactly:

$$\text{diameter case (IV.13) \quad or \quad (IV.14 + IV.12.1).}$$

18 Hilbert and Book Zero Components

18.1 Plane separation

The entire generic proof is organized by same-side and opposite-side relations with respect to the diagonal BD and the antipodal diameter AE .

The formal proof must transport these relations through betweenness and through the antipodal extension $A - K - E$.

18.2 Betweenness and trichotomy

The key order configurations are

$$A - K - E, \quad A - Y - E, \quad B - Y - D.$$

The last one is not assumed; it is derived by trichotomy and metric contradiction.

18.3 Segment comparison

The proof that exterior orders on B, Y, D are impossible uses strict segment comparison: an exterior point on the chord carrier would be farther from the centre than a circle radius, contradicting the fact that Y is inside the antipodal diameter.

18.4 Book Zero supplements

The final conclusion is not an arithmetic equality

$$\alpha + \beta = \pi.$$

Instead two explicit supplement configurations are compared by

```
bookZero_43_supplements
```

This is the synthetic mechanism behind the substitution of congruent inscribed angles into supplementary relations.

19 Foundational Boundary

IV.15 introduces no new axiom.

The proof uses:

- circle existence already packaged in `HilbertConcyclic4`;
- incidence and line uniqueness;
- plane separation;
- order trichotomy and betweenness transitivity;
- segment congruence and strict comparison;
- the established numbered theorems IV.12.1, IV.13, and IV.14;
- Book Zero supplement transport.

No numerical angle measure, coordinate geometry, or continuity argument is introduced.

20 Classical and Formal Dependency Structures

The source proof is compact:

$$\begin{array}{c}
 ABCD \text{ cyclic, } A, C \text{ opposite sides of } BD \\
 \downarrow \\
 \text{centre } K \\
 \downarrow \\
 \left\{ \begin{array}{l} K \in BD \implies \text{IV.13,} \\ K \notin BD \implies \text{choose antipode } E \end{array} \right. \\
 \downarrow \\
 \text{IV.14 on } A, B, E, D \\
 \downarrow \\
 \angle BAD + \angle BED = \pi \\
 \downarrow \\
 \text{IV.12.1: } \angle BED \cong \angle BCD \\
 \downarrow \\
 \angle BAD + \angle BCD = \pi.
 \end{array}$$

The formal generic branch expands the middle:

$$\begin{array}{c}
 C \sim K \text{ across } BD \\
 \downarrow \\
 A - K - E \\
 \downarrow \\
 A \mid E \text{ across } BD \\
 \downarrow \\
 Y = AE \cap BD \\
 \downarrow \\
 Y = K \text{ or } KY < KA \\
 \downarrow \\
 \text{exclude } Y - B - D \text{ and } B - D - Y \\
 \downarrow \\
 B - Y - D \\
 \downarrow \\
 B \mid D \text{ across diameter } AE \\
 \downarrow \\
 \text{IV.14} \\
 \downarrow \\
 E \sim C \text{ across } BD \\
 \downarrow \\
 \text{IV.12.1} \\
 \downarrow \\
 \text{Book Zero supplement transport.}
 \end{array}$$

The module dependency graph is

```

HilbertInterfaceIV
|
+--> Proposition4_12
|

```

```

+--> Proposition4_13
|
+--> Proposition4_14
    |
    v
    Proposition4_15

```

More precisely, `Proposition4_15.lean` imports all three numbered predecessors because each has a distinct mathematical role.

21 Diagrammatic Audit

Four figures are useful for IV.15.

1. The source cyclic quadrilateral and diagonal BD .
2. The special case $K \in BD$, where the diagonal is a diameter.
3. The generic antipodal construction $A - K - E$ with the crossing point Y .
4. The relabelled symmetry branch that Lean handles explicitly rather than by a verbal “without loss of generality”.

The difficult order proof excluding exterior positions of Y is better explained in text and inequalities than by adding yet another crowded figure.

22 Conclusion

IV.15 is the first proposition in the Forder circle block where the formal proof reveals the full cost of a classical symmetry argument.

The centre-on-diagonal case is short and uses IV.13. The generic case introduces an antipodal diameter, proves that the original diagonal crosses that diameter internally, applies IV.14, and then uses IV.12.1 to replace the auxiliary inscribed angle by the target angle. The final substitution is performed through explicit Book Zero supplement witnesses.

Most importantly, the alternative placement of the centre is not dismissed as “symmetric”. The Lean proof performs the cyclic relabelling and transports the supplement back to the original orientation.

The result is a production theorem whose public statement is simple but whose documentation exposes the actual incidence, order, congruence, and symmetry machinery hidden in the printed diagram.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.